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X-AVDT: Audio-Visual Cross-Attention for Robust Deepfake Detection

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Abstract

The surge of highly realistic synthetic videos produced by contemporary generative systems has significantly increased the risk of malicious use, challenging both humans and existing detectors. Against this backdrop, we take a generator-side view and observe that internal cross-attention mechanisms in these models encode fine-grained speech–motion alignment, offering useful correspondence cues for forgery detection. Building on this insight, we propose ***X-AVDT****, a robust and generalizable deepfake detector that probes generator-internal audio-visual signals accessed via DDIM inversion to expose these cues. X-AVDT extracts two complementary signals: (i) a video composite capturing inversion-induced discrepancies, and (ii) audio–visual cross-attention feature reflecting modality alignment enforced during generation. To enable faithful, cross-generator evaluation, we further introduce* ***MMDF****, a new multi-modal deepfake dataset spanning diverse manipulation types and rapidly evolving synthesis paradigms, including GANs, diffusion, and flow-matching. Extensive experiments demonstrate that X-AVDT achieves leading performance on MMDF and generalizes strongly to external benchmarks and unseen generators, outperforming existing methods with accuracy improved by* ***+13.1%****. Our findings highlight the importance of leveraging internal audio–visual consistency cues for robustness to future generators in deepfake detection. Code is available at X-AVDT.*

1. Introduction

Deepfakes, synthetic or edited portrait videos that manipulate identity, speech, or motion, have become feasible with advances in generative video models. These models have evolved from Generative Adversarial Networks (GANs) [27] to diffusion-based models [34, 71], which push fidelity to unprecedented levels. Recent systems [19, 33, 87] can synthesize photorealistic digital humans from minimal input, lowering production costs for creative media and assistants. However, when misused, the same advances heighten societal and security risks, including targeted disinformation, real-time impersonation, identity theft, and financial fraud [1–3, 17, 23, 28]. These concerns have motivated the research community to develop deepfake detection, which aims to reliably authenticate media under rapidly evolving generators.

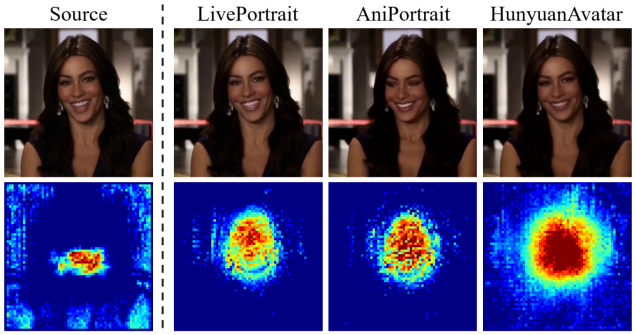


Figure 1. **Temporally averaged cross-attention maps.** For each video, we extract audio–visual cross-attention during DDIM inversion and average the maps over all frames to obtain a single heatmap. Real vs. fake samples exhibit consistent disparities.

Diffusion models have become central to recent facial forgery generation. Many video generators employ cross-attention [83] to condition visual features on external signals such as text, motion, or audio [7, 15, 84]. Among these external signals, audio is especially useful as it provides frame synchronous, densely informative supervision aligned with facial dynamics. In audio-driven diffusion models, this conditioning often takes the form of audio–visual cross-attention, which ties phonetic content to facial motion and expressiveness. Such architectures are explicitly designed to promote audio-visual alignment via cross-attention in the diffusion U-Net, making their internal features a natural source of correspondence cues for deepfake detection. Figure 1 presents examples of extracted audio-visual cross attention features, using Hallo [90], an audio-driven talking head diffusion model pretrained on a large corpus of speech videos. It is observed that similar attention patterns recur across different generator frameworks. This indicates that internal audio–visual cross-attention features from diffusion models provide a robust, generator-agnostic discriminative signal for deepfake detection, reinforcing this generality.

Dataset	Modality	Method			Model		
		FS	RE	TH	GAN	DF	FM
Celeb-DF [49]	V	✓	✗	✗	✓	✗	✗
DF-Platter [61]	V	✓	✓	✗	✓	✗	✗
KoDF [41]	AV	✓	✓	✓	✓	✗	✗
FaceForensics++ [72]	AV	✓	✓	✗	✓	✗	✗
DFDC [22]	AV	✓	✗	✗	✓	✗	✗
FakeAVCeleb [39]	AV	✓	✗	✓	✓	✗	✗
AV-Deepfake1M [10]	AV	✗	✗	✓	✓	✗	✗
MMDF (Ours)	AV	✓	✓	✓	✓	✓	✓

Table 1. **Comparison of deepfake video datasets.** FS:Face swapping, RE: Self-reenactment, TH: Talking-head generation, DF: Diffusion model, FM: Flow-matching.

Building on this observation, we propose **X-AVDT**, an **A**udio-**V**isual **Cross**-Attention framework for robust **D**eepfake de**T**ection. Our framework leverages the cross-modal interaction of videos by utilizing fine-grained audio-visual diffusion features to generalize across manipulation types and synthesis models. To extract internal signals, we employ an inversion scheme that maps input videos into the diffusion model's latent space and reconstructs them under the model prior. We further incorporate a latent noise map, reconstructed video, and input-reconstruction residual as complementary spatial cues, motivated by findings that pretrained diffusion models more faithfully reconstruct diffusion-generated content than real content [11, 86]. While fully synthetic videos often expose global inconsistencies, face-centric manipulations are confined to the facial region, preserve identity, and yield subtle artifacts that are easily obscured, thereby challenging residual-only detectors [11, 86]. By augmenting inversion-based discrepancies with audio-visual cross-attention and fusing them into a unified representation, X-AVDT provides complementary global and localized evidence that can improve detection reliability.

Existing datasets [22, 45, 49] are largely composed of earlier GAN-generated forgeries, offering limited coverage of contemporary models and manipulation types. As a result, they fail to capture the diversity and realism of continuously updated diffusion or flow-based methods, constraining progress toward building detectors that generalize beyond legacy benchmarks. To further facilitate robust detection of rapidly evolving deepfakes, we introduce **MMDF**, a curated **M**ulti-modal, **M**ulti-generator **D**eep**F**ake dataset. It is a high-quality dataset of paired real videos and corresponding fakes generated by a diverse suite of recent synthesis models. MMDF is the first dataset to cover contemporary diffusion (both U-Net [71] based and transformer [64] based) and flow-matching [52] generators, and it includes audio–visual pairs. The dataset also spans manipulation paradigms such as talking-head generation [15, 19], self-reenactment [30, 91], and face swapping [33], making it suitable for real-world, unconstrained deepfake detection. A comparison of MMDF with prior deepfake datasets is provided in Table 1.

2. Related Work

Deepfake Video Generation. The synthesis and editing of human faces in video, with high realism and temporal coherence, have been extensively studied. Early innovations in generative modeling were driven by GANs [27] with gains in stability, resolution, and controllability [37, 38, 68], along with basic conditional setups [36, 60, 98]. More recently, diffusion and flow-matching models [15, 34, 71] have shown remarkable success in portrait video manipulation for deepfakes. In parallel, control signals such as audio [19, 87], textual prompts [7, 88], facial landmarks [84, 94], dense motion flow [94], and 3D face priors [51, 65] have significantly enhanced the realism of synthetic videos. Moreover, current manipulation techniques include talking-head generation [15, 19], face reenactment [30, 91], face swapping [33], lip-sync/dubbing [66, 96], and appearance editing [9].

Artifact-based Deepfake Detection. Detection methods have evolved alongside deepfake video generation. Early attempts relied on hand-crafted forensic features, including blink detection [48], facial warping artifacts [47], or color inconsistencies [43, 59]. CNN-based classifiers trained on real and fake examples became dominant. Prior studies have demonstrated the effectiveness of deep learning for detecting subtle synthesis artifacts [4, 29, 46, 67, 72, 73, 77, 85]. Frequency-domain and fingerprint-based detectors further improved generalization across different GAN architectures [26, 58]. However, these methods often struggle to generalize beyond their training datasets and remain vulnerable to newer, more sophisticated forgeries.

Generalizable Deepfake Detection. A principal goal is generalization across unseen generators, manipulation types, and domains. To this end, recent work investigates the semantic characteristics of generated images. DIRE [86] proposes a reconstruction-based detector grounded in the hypothesis that images produced by diffusion models can be more accurately reconstructed by a pretrained diffusion model than real images. DRCT [13] extends this idea by synthesizing hard examples via diffusion reconstruction and applying contrastive learning to the resulting residuals. FakeInversion [11] leverages features obtained via latent inversion of Stable Diffusion [71]. These methods are built on prior observations that CLIP [69] embeddings can be predictive of image authenticity [62, 75].

Audio-Visual Inconsistencies. Several studies [24, 40, 78, 97] adopt a late-fusion design in which RGB images and audio are encoded separately and combined only at the classification head. While this preserves strong per-modality features, the resulting embeddings occupy different latent spaces and are not directly aligned across modalities. Other

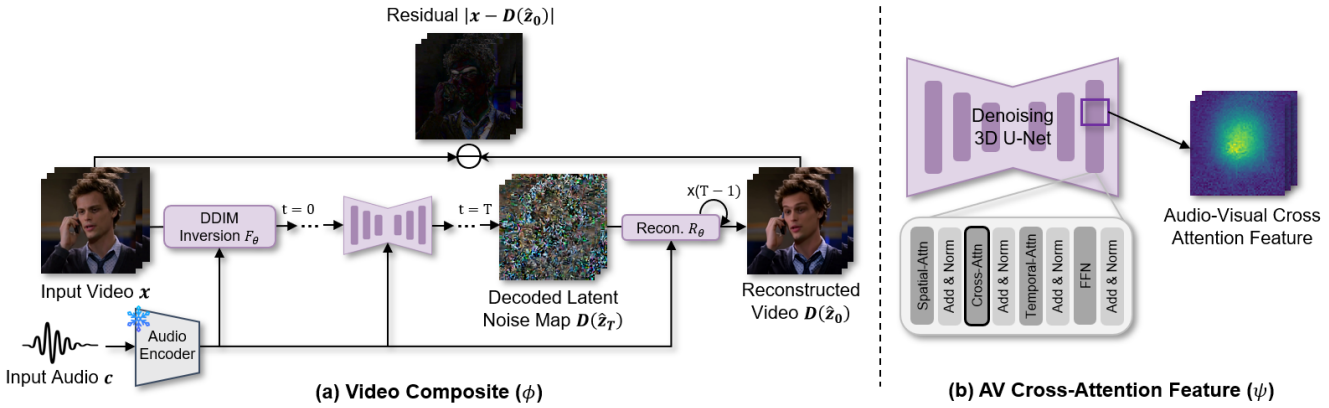


Figure 2. **Input representations with complementary features.** (a) video composite ϕ is obtained from video x and audio c by running DDIM inversion and reconstruction, decoding both the noisy and clean latents, and computing the residual. We then concatenate four components channel-wise: x, $D(\hat{z}_T)$, $D(\hat{z}_0)$, and $|x - D(\hat{z}_0)|$. (b) AV cross-attention feature ψ is extracted during DDIM inversion from the diffusion U-Net and summarized as a frame-aligned tensor. These complementary cues (a) and (b) capture appearance information and modality alignment, respectively. For clarity, all visual elements shown ($D(\hat{z}_T)$, $D(\hat{z}_0)$, and $|x - D(\hat{z}_0)|$) are decoded images.

approaches [32, 50, 63] learn audio–visual representations in a self-supervised way, implicitly fusing the modalities by pulling their embeddings together. Such implicit fusion can miss fine-grained semantic misalignment and offers limited interpretability of the cross-modal evidence. In contrast, we focus on internal features from large generative models as explicit, interpretable signals of persistent audio-visual inconsistency.

3. Preliminaries

Diffusion models [21, 34, 71] are probabilistic generative models that learn a data distribution by progressively denoising samples. They consist of two complementary processes. In the forward process, Gaussian noise is gradually injected into a clean image x_0, given by

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (1)$$

which maps x_0 to increasingly noisy latents x_t, where $t = 0, \ldots, T$ and $\bar{\alpha}_t = \prod_{k=1}^{t} \alpha_k$. The reverse process starts from a noisy sample x_T and iteratively produces cleaner states conditioned on an auxiliary vector c (e.g., audio), using a noise predictor $\epsilon_\theta(x_t, t, c)$. Given the current estimate of the clean image,

$$\hat{x}_0(x_t, t, c) = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\, \epsilon_\theta(x_t, t, c)}{\sqrt{\bar{\alpha}_t}}, \quad (2)$$

the conditional reverse update can be written as

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\, \hat{x}_0(x_t, t, c) + \sqrt{1 - \bar{\alpha}_{t-1}}\, \epsilon_\theta(x_t, t, c). \quad (3)$$

In this work, we rely on a pre-trained audio-conditioned Latent Diffusion Model (LDM) [71], where the diffusion process operates in the latent space of a VAE: $z_0 = E(x_0)$ and $x_0 \approx D(z_0)$. The denoiser is a 3D U-Net conditioned on external signals and composed of residual blocks, spatial self-attention, temporal self-attention, and cross-attention. Let f_n^l be the video features at layer l and video frame n, projected to queries q_n^l, keys k_n^l, and values v_n^l. The attention output is given by

$$\tilde{f}_n^l = A_n^l\, v_n^l, \qquad A_n^l = \mathrm{Softmax}\left(\frac{q_n^l (k_n^l)^\top}{\sqrt{d}}\right), \quad (4)$$

where d is the query/key embedding dimension. This enables the model to capture global dependencies across space and time in the video. In parallel, cross-attention is computed between video queries and the conditioning audio embedding, thereby guiding the denoising process toward the target reconstruction.

4. Method

4.1. Problem Definition

We train a binary deepfake detector on audio–visual pairs $\mathcal{D}_{\text{train}} = \{(x_m, c_m, y_m)\}_{m=1}^{M}$, where each example consists of a face video clip x_m, its paired audio condition c_m, and a label $y_m \in \{0, 1\}$ that indicates whether the video is real or fake. The training set contains M videos. From each input pair, an audio-conditioned LDM extracts two types of features, video composite $\phi(x, c)$ and an AV cross-attention feature $\psi(x, c)$, as illustrated in Figure 2 and discussed in Sec. 4.2. We fuse ϕ and ψ and train a detector G_θ to estimate $p_\theta(y = 1 \mid x, c) = \sigma(G_\theta(\cdot))$ by optimizing a weighted sum of a binary cross-entropy term and a metric-learning term, which will be defined in Sec. 4.3.

4.2. Input Representation

4.2.1. Diffusion Inversion & Reconstruction

As shown in Figure 2a, given a video $x \in \mathbb{R}^{N \times 3 \times H \times W}$, where N denotes the number of frames in the clip and H and W denote the height and width, respectively, and a

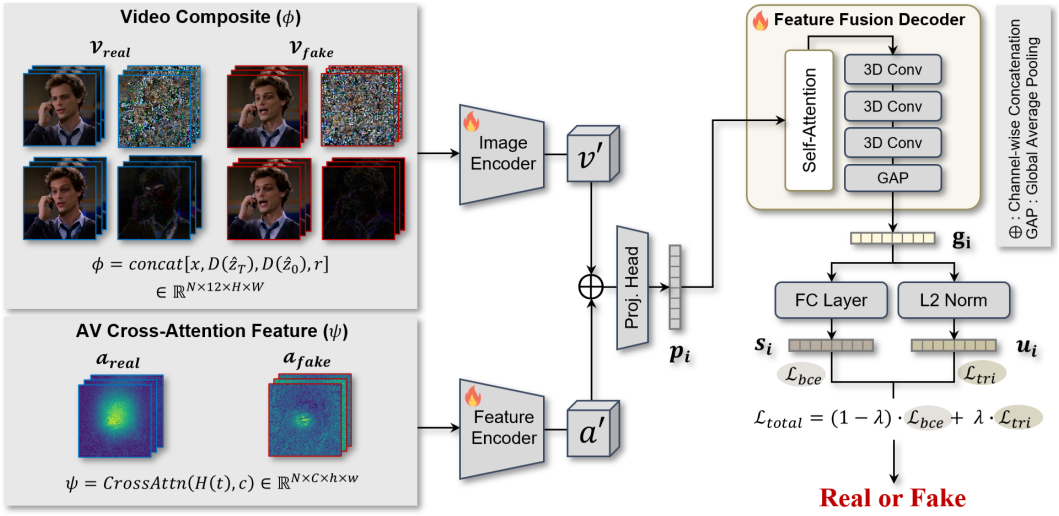


Figure 3. **The overall framework of X-AVDT.** From each audio-visual pair, we form two inputs ϕ and ψ. Two 3D encoders map them to features that are concatenated and passed through the Feature Fusion Decoder to produce a fused feature. A classification head outputs the real/fake score, while an embedding head is trained with a triplet objective to improve robustness.

paired audio condition c, we operate in the VAE latent space with encoder E and decoder D. We first encode the input video frame to $z_0 = E(x)$ while conditioning on the wav2vec 2.0 [6] audio embedding c (omitted in the figure for brevity). We then obtain the corresponding latent noise map using the DDIM inversion process $\hat{z}_T = F_\theta(z_0, c)$. Subsequently, starting from $\hat{z}_T$, we run the conditional reverse diffusion to obtain a clean latent $\hat{z}_0 = R_\theta(\hat{z}_T, c)$. We then decode $\hat{z}_0$ to the pixel space to obtain the reconstructed video $D(\hat{z}_0)$ and compute the residual between the input and the reconstruction $r = |x - D(\hat{z}_0)|$.

To construct the video composite ϕ for the detector, we use an input formed by channel-wise concatenation of the image x, the decoded latent DDIM noise map $D(\hat{z}_T)$, the image $D(\hat{z}_0)$ reconstructed from the reverse DDIM process, and the reconstruction residual $r = |x - D(\hat{z}_0)|$:

$$\phi(x, c) = \text{concat}\big[\, x,\ D(\hat{z}_T),\ D(\hat{z}_0),\ r \,\big] \in \mathbb{R}^{N\times 12\times H\times W}, \tag{5}$$

Because DDIM inversion uses a finite number of steps, the mismatch after a forward-reverse pass reflects discretization error, whereas manipulated samples tend to produce smaller discrepancies and thus higher likelihood of forgery under the diffusion model [11]. We use the pattern of this gap as an inversion-induced discrepancy measure for detecting manipulation.

4.2.2. Audio-Visual Cross-Attention Feature

As illustrated in Figure 2b, DDIM inversion is performed with a 3D U-Net composed of multiple blocks organized into down, mid and up stages. Each block contains an audio-visual cross-attention layer in which video hidden states serve as queries while the audio encoder's hidden states provide keys and values. We extract the cross-attention from an up block at a chosen diffusion timestep t, conditioned on the input audio. This attention is taken from the same conditioned LDM used to construct the video composite ϕ.

Let $H(t)$ be the 3D U-Net hidden state at timestep t. Using the attention output projection, we aggregate the multi-head outputs, reduce the head dimension to C channels, and reshape the result into the per-frame latent grid, yielding a temporally aligned feature $\psi(x, c)$, which is defined as:

$$\psi(x, c) = \text{CrossAttn}(H(t),\, c) \in \mathbb{R}^{N\times C\times h\times w}, \tag{6}$$

Here, $h \times w$ is the latent-space resolution (e.g., 64×64 for 512×512 inputs with an 8x downsampling). In our setup, we extract the attention from the last up block at timestep $t = 24$ and reshape it to a per-frame. The resulting ψ provides a compact, temporally aligned descriptor of audio–visual correspondence that the detector can exploit to distinguish authentic from manipulated content. Because it captures speech–motion synchrony enforced by the denoiser instead of appearance alone, it is less sensitive to purely visual artifacts and thus offers a complementary, model-internal cue that improves robustness. Ablation on input representations and attention features appear in Sec. 6.4.

4.3. Detector Architecture

Figure 3 presents the X-AVDT framework. X-AVDT pairs two signals extracted from different parts of the conditioned LDM pipeline to improve robustness. Accordingly, the detector takes two inputs. Two 3D encoders, E_v for ϕ and E_a for ψ, produce feature volumes that are aligned in space and time and then fused by a Feature Fusion Decoder (FFD) to yield a logit and an embedding:

$$\mathbf{v}' = E_v(\phi), \qquad \mathbf{a}' = E_a(\psi). \tag{7}$$

The tensors $\mathbf{v}'$ and $\mathbf{a}'$ are concatenated along the channel dimension and projected to a shared embedding with a 1×1 convolution to obtain $\mathbf{p}_i$. The FFD then applies a self-attention layer over spatial tokens, followed by a series of L 3D ResNeXt [89] layers. Then global average pooling (GAP) produces a fused feature vector $\mathbf{g}_i$ for the i-th sample:

$$\mathbf{p}_i = \mathrm{Proj}(\mathrm{concat}[\mathbf{v}'_i, \mathbf{a}'_i]), \tag{8}$$

$$\mathbf{g}_i = \mathrm{GAP}\big(\mathrm{Conv3D}^L\big(\mathrm{SelfAttn}(\mathbf{p}_i)\big)\big). \tag{9}$$

From $\mathbf{g}_i$ we form two branches: a fully connected layer maps $\mathbf{g}_i$ to a scalar logit s_i for the binary classification loss, and an embedding head outputs an ℓ_2-normalized vector $\mathbf{u}^{(i)}$ for metric learning. With ground-truth labels $y_i \in \{0, 1\}$ (real=0, fake=1) and sigmoid $\sigma(\cdot)$, the binary cross-entropy loss over a mini-batch of size B is defined as follows:

$$\mathcal{L}_{\text{bce}} = -\frac{1}{B}\sum_{i=1}^{B}\Big[y_i \log\sigma(s_i) + (1-y_i)\log\big(1-\sigma(s_i)\big)\Big]. \tag{10}$$

Let $\mathbf{u}_a^{(i)}$, $\mathbf{u}_p^{(i)}$, and $\mathbf{u}_n^{(i)}$ denote anchor, positive, and negative embeddings for the i-th triplet, respectively. It tightens same class embeddings and separates different classes. Using the squared ℓ_2 distance and a margin $m > 0$, the triplet loss is

$$\mathcal{L}_{\text{tri}} = \frac{1}{B}\sum_{i=1}^{B}\max\big(0,\ \big\|\mathbf{u}_a^{(i)} - \mathbf{u}_p^{(i)}\big\|_2^2 - \big\|\mathbf{u}_a^{(i)} - \mathbf{u}_n^{(i)}\big\|_2^2 + m\big). \tag{11}$$

The overall objective minimizes a weighted sum of the two terms, controlled by a balancing parameter $\lambda \in [0, 1]$:

$$\mathcal{L}_{\text{total}} = (1-\lambda)\,\mathcal{L}_{\text{bce}} + \lambda\,\mathcal{L}_{\text{tri}}. \tag{12}$$

This joint objective allows the model to learn discriminative and mutually informative representations that generalize across manipulation patterns. Further analysis is provided in Sec. 6.4.

5. Dataset

The MMDF dataset was constructed from curated real clips paired with their corresponding manipulated versions, resulting in a collection of 28.8k clips with a total duration of 41.67 hours. Using Hallo3 dataset [20] as our source, we applied each video clip duration, resolution, and face-presence filters with MediaPipe [56] to retain single-person, frontal-to-quarter views with stable lip motion and speech. We removed clips with scene cuts, strong camera motion, excessive facial motion, or long side facing poses. These videos contain individuals of diverse ages, ethnicities, and genders in close-up shots across varied indoor and outdoor backgrounds. MMDF covers three manipulation types, such as talking-head generation, self-reenactment, and face swapping, produced using GAN, diffusion, and flow-matching generators. Comparative networks were trained on a diverse set of data obtained from commonly used generators and evaluated on unseen ones to assess cross-generator generalization. See Table 2 for details.

Split	Generator	Model	Method	#Clips (real/fake)
Train	Hallo2 [19]	Diffusion	Talking-Head	4k/4k
	LivePortrait [30]	GAN	Self-Reenactment	4k/4k
	FaceAdapter [33]	Diffusion	Face Swapping	4k/4k
Test	HunyuanAvatar [15]	Flow Matching	Talking-Head	0.8k/0.8k
	MegActor-Σ [91]	Diffusion Transformer	Self-Reenactment	0.8k/0.8k
	AniPortrait [87]	Diffusion	Talking-Head	0.8k/0.8k

Table 2. **Composition of the dataset.** This coverage better reflects current synthesis trends than outdated datasets that focus primarily on GANs, and permits assessing cross-generator generalization.

Dataset	Sync-C ↑	Sync-D ↓	LPIPS ↓	FVD ↓	HFAR ↑
FaceForensics++ [72]	3.32	11.06	0.27	370.23	0.22
FakeAVCeleb [39]	5.87	8.38	0.19	170.61	0.34
MMDF (Ours)	**7.36**	**7.35**	**0.07**	**121.39**	**0.41**

Table 3. **Audio-visual quality of manipulated videos.** Sync-C, Sync-D, LPIPS, and FVD are computational metrics, whereas HFAR denotes the human false acceptance rate measured in a user study.

To quantify the realism of the manipulated videos, we report several metrics values. As shown in Table 3, MMDF exhibits stronger audio–visual synchronization and more favorable perceptual and video statistics than FaceForensics++ [72] and FakeAVCeleb [39], indicating temporally coherent, high-quality manipulations suitable for the detection task. In addition, we report Human False Acceptance Rate (HFAR), defined as the proportion of fake clips judged to be real by human evaluators (i.e., False Positive Rate, FPR). A higher HFAR indicates that humans are more likely to perceive fake clips as real, suggesting that the dataset contains more challenging and realistic negative samples. Metric definitions are provided in Sec. 6.1 and MMDF details are in the supplementary material.

6. Experiments

6.1. Implementation Details

X-AVDT is implemented and trained with the same configuration across all evaluation datasets. Training requires 14 hours on a single NVIDIA RTX 3090 GPU.

Architecture. Hallo [90] is employed as our audio-conditioned latent diffusion backbone, initialized from Stable Diffusion [71]. While we choose Hallo because it provides high-fidelity synthesis and rich internal audio–visual signals that our detector can exploit, other choices can also be possible [16, 53, 87]. See the supplementary material for details. Audio conditioning is provided by wav2vec 2.0 [6] features projected to the U-Net cross-attention embedding

Model	AniPortrait [87]				MegActor-Σ [91]				HunyuanAvatar [15]				Average			
	AUROC	AP	Acc@EER	Acc	AUROC	AP	Acc@EER	Acc	AUROC	AP	Acc@EER	Acc	AUROC	AP	Acc@EER	Acc
Official-pretrained																
LipForensics [31]	79.49	81.45	73.25	74.50	72.17	73.56	65.85	67.11	63.66	56.50	59.91	60.66	74.24	74.54	71.91	72.38
RealForensics [32]	86.31	85.23	78.65	77.56	62.93	61.60	58.79	58.71	64.90	60.30	61.40	60.85	77.65	75.71	72.15	65.70
AVAD [24]	73.25	74.98	66.05	67.49	50.87	56.96	51.26	51.18	75.37	74.54	74.14	74.39	67.47	67.29	64.75	65.29
FACTOR [70]	94.21	93.54	86.29	86.55	75.44	72.83	68.82	69.33	36.69	46.18	38.27	56.15	68.78	70.85	64.46	70.68
LipFD [54]	52.55	50.60	51.74	51.90	51.45	49.89	50.80	50.93	60.38	53.85	57.78	58.30	57.64	51.83	55.37	55.82
AVH-Align [78]	74.88	73.94	68.82	70.03	51.20	47.81	50.86	50.93	34.53	39.96	37.50	36.91	50.34	50.95	49.89	49.98
MMDF-retrained (cross-dataset evaluation)																
RealForensics [32]	97.47	97.22	90.99	90.28	**97.60**	**96.70**	**92.68**	**92.45**	74.89	72.09	68.38	61.12	92.42	91.39	84.01	81.28
LipFD [54]	49.59	50.26	49.79	49.75	50.32	50.82	50.14	50.18	48.39	50.26	49.10	48.96	53.75	51.29	52.65	52.87
AVH-Align [78]	98.94	98.53	95.94	96.54	67.92	62.36	62.24	63.38	75.92	65.87	69.70	70.94	81.44	76.52	75.59	76.76
Human Evaluation	–	–	–	83.75	–	–	–	74.17	–	–	–	58.33	–	–	–	71.88
X-AVDT (Ours)	**99.10**	**98.89**	**96.54**	**97.05**	90.17	88.05	83.11	84.52	**97.79**	**97.44**	**97.69**	**97.91**	**95.29**	**94.03**	**91.15**	**91.98**

Table 4. **Quantitative comparison on the MMDF dataset.** Detectors are evaluated using the official pretrained checkpoint (first panel) and after retrained on the MMDF training set (Hallo2 [19], LivePortrait [30], and FaceAdapter [33]) (second panel). Best in **bold**; second-best underlined. *Note: FACTOR is a zero-shot method, while AVAD and AVH-Align are unsupervised methods.

Model	FakeAVCeleb [39]				FaceForensics++ [72]			
	AUROC	AP	Acc@EER	Acc	AUROC	AP	Acc@EER	Acc
Official pretrained								
LipForensics [31]	98.40	98.37	95.00	95.80	98.14†	97.93†	97.00†	98.50†
RealForensics [32]	95.80	96.20	85.99	85.46	99.47†	99.42†	95.49†	95.22†
AVAD [24]	74.96	75.88	66.00	66.79	55.38	53.92	51.50	52.16
FACTOR [70]	88.44	88.22	76.00	80.00	76.33	76.08	68.50	71.00
LipFD [54]	73.17	64.39	66.36	67.16	52.84†	51.01†	50.96†	59.31†
AVH-Align [78]	93.52	93.48	83.00	83.90	37.07	40.43	40.50	41.62
MMDF-trained (cross-dataset evaluation)								
RealForensics [32]	83.67	85.56	71.42	71.42	88.85	87.65	79.39	78.54
LipFD [54]	53.59	51.09	52.37	53.67	52.92	50.21	52.08	51.89
AVH-Align [78]	54.20	53.74	52.00	52.80	36.66	40.03	37.00	39.64
Human Evaluation	–	–	–	78.75	–	–	–	71.25
X-AVDT (Ours)	**99.69**	**99.74**	**97.85**	**98.65**	**89.55**	**89.17**	**87.55**	**89.77**

Table 5. **Quantitative comparison on the benchmark dataset.** Detectors are trained on the MMDF training set and evaluated on FakeAVCeleb and FaceForensics++, respectively. †Indicates that the corresponding benchmark was used during the method's original training (train–test overlap).

dimension. We do not use classifier-free guidance during either DDIM inversion or reconstruction to preserve bijectivity and conditioning fidelity. Unless otherwise stated, we extract the audio–visual cross-attention from the last up block of the U-Net at diffusion timestep $t = 24$. Both the image encoder $E_v(\cdot)$ and the feature encoder $E_a(\cdot)$ use 3D ResNeXt [89]. The FFD uses an L-layer 3D ResNeXt stack. We fix $L = 3$ in Eq. (8). In our setup, the video composite ϕ has 12 channels (x, $D(\hat{z}_T)$, $D(\hat{z}_0)$, residual), and the AV cross-attention feature ψ uses $C = 320$ channels after collapsing the multi-head dimension and is reshaped with a latent resolution of 64×64.

Training Details. We train the detector for 2 epochs on frames of size 512×512 using AdamW [55] with a learning rate of 1×10^{-4}, weight decay of 0.05, and a batch size of 8. For the triplet loss Eq. (11), we set the margin m to 0.3, and the balancing parameter λ in the overall objective Eq. (12) is also set to 0.3.

Automatic Metrics. We report the values measured by four detection metrics, AUROC, Average Precision (AP), Accuracy at Equal Error Rate (Acc@EER), and Accuracy. We obtain Acc@EER by computing the EER threshold from the ROC curve and evaluating accuracy at that threshold. To assess the proposed MMDF dataset, we also report values measured by generation quality metrics including Lip Sync scores (Sync-C and Sync-D) from SyncNet for lip-speech alignment [18], LPIPS for perceptual distance [95], and Fréchet Video Distance (FVD) for distributional distance between generated and ground-truth videos [82]. Lower is better for LPIPS and FVD, higher Sync-C and lower Sync-D indicate better synchronization.

Baselines. We evaluated X-AVDT against open-source baselines, including the video-only LipForensics [31], and audio–visual methods RealForensics [32], AVAD [24], LipFD [54], FACTOR [70], and AVH-Align [78]. We used two variants: (i) the official checkpoint and (ii) models retrained from scratch on our training sets using the official code. This protocol enables a fair cross-dataset transfer comparison against our MMDF-trained model. Further details are in the supplementary material.

6.2. Comparison with State-of-the-art

6.2.1. Comparison on MMDF Dataset

Table 4 comprises two panels. *Official-pretrained* reports results from publicly released checkpoints of prior methods, and *MMDF-retrained* reports the same baselines retrained on our MMDF for cross-dataset evaluation. Baselines are evaluated in both settings, whereas X-AVDT values are reported in the *MMDF-retrained* panel. As the training code for LipForensics [54] and AVAD [24] is unavailable, we report results obtained by their official pretrained model only. The pretrained baselines presented in the first panel tended to overfit to earlier synthesis methods and showed limited domain adaptation capability, resulting in poor transfer abil-

	Cross-Attention			Temporal-Attention			Spatial-Attention		
Timestep t	AUROC	AP	Acc@EER	AUROC	AP	Acc@EER	AUROC	AP	Acc@EER
$t = 24$	**91.56**	86.90	**81.51**	83.92	**87.08**	75.01	64.57	68.00	61.82
$t = 249$	81.30	69.52	77.13	68.25	71.94	64.10	57.42	52.00	56.04
$t = 499$	68.11	57.97	67.44	66.29	66.29	59.30	52.38	53.14	50.02

Table 6. **Ablation results for different attention features and diffusion timesteps.** As the diffusion timestep t grows, the latent becomes noisier and conditioning weakens. Results at $t \in \{24, 249, 499\}$ show that cross-attention performs best, and performance degrades as t increases.

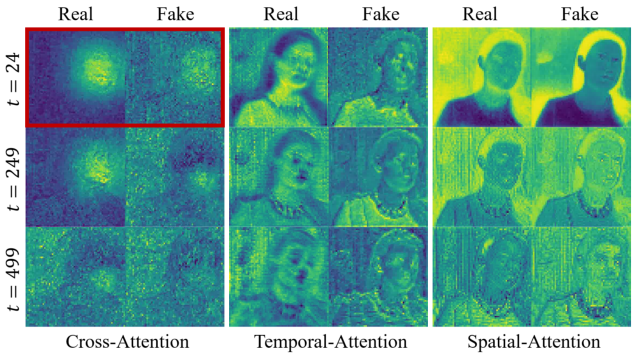


Figure 4. **Comparison of attention features across diffusion timesteps.** Red box denote the configuration used in our method.

ity to unseen manipulations and a persistent generalization gap across datasets. As shown in the bottom row, X-AVDT achieved the highest average AUROC of 95.29, exceeding the performance of the strongest retrained baseline (RealForensics [32] 92.42). The margin over prior methods remained broadly consistent across generators, resulting in a clear overall lead despite MMDF retraining.

6.2.2. Comparison on Benchmark Dataset

We further evaluate X-AVDT on the GAN-based benchmarks FakeAVCeleb [39] and FaceForensics++ [72], following the same setup as in Table 4. As summarized in Tables 4 and 5, models trained on MMDF transfer well to these benchmarks. Notably, several pretrained detectors are marked with †, indicating train-test overlap. Even under this favorable condition for the baselines, X-AVDT achieved the best scores on both benchmarks with an AUROC of 99.69 on FakeAVCeleb and 89.55 on FaceForensics++.

6.3. Human Evaluation

For human evaluation studies, we use two metrics: Human Evaluation (HE) in Table 4 and Table 5, and HFAR in Table 3. Both HE and HFAR were obtained from the same user study with 24 participants (balanced in gender, aged between their 20s and 30s), where each participant viewed the videos with audio and answered whether it was real or fake. Across datasets, HE was consistently lower than our detector, indicating the task is challenging for humans, whereas our model remains robust.

Method	AUROC	AP	Acc@EER
(a) Ablation on Input Representations			
w/o AV Cross-Attn (ψ)	88.22	87.25	83.70
w/o Video Composite (ϕ)	90.21	90.57	84.32
w/o Residual ($\lvert x - D(\hat{z}_0)\rvert$)	93.82	92.25	89.00
(b) Ablation on Loss Design			
w/o $\mathcal{L}_{\text{tri}}$	92.64	92.26	86.32
X-AVDT (full)	**95.29**	**94.03**	**91.15**

Table 7. **Ablation on input representations and loss design.** (a) removing any of the input representation degrades performance, (b) adding the triplet term improves the results across all metrics.

6.4. Ablation Study

Choice of Attention Type and Timestep. We analyze how the choice of attention feature and the diffusion timestep affect detection. Table 6 shows that, among attention features extracted from the 3D U-Net in Hallo [90], the audio-conditioned cross-attention was consistently the most informative, as it explicitly regularizes the representation toward audio–visual alignment. Figure 4 further illustrates that temporal coherence and spatial appearance tend to capture global motion and pose changes, making them less discriminative. By contrast, cross-attention highlights articulators while suppressing background and is inherently less sensitive to scene changes. Moreover, features from earlier diffusion steps were more discriminative than those from later steps, because early denoising retains stronger conditioning signals before texture refinement dominates, leaving modality-consistency cues less degraded.

Complementary Effect of Input Representations. The video composite ϕ encodes inversion-induced discrepancies, while the AV cross-attention feature ψ provides cross-modal consistency cues derived from the diffusion model's internal alignment. Motivated by their complementary nature, we combine ϕ and ψ and assess their contributions via an ablation study that removes each component individually. The combined model yields the highest overall performance, indicating that the two representations reinforce each other. Table 7a corroborates this with quantitative results across all three metrics.

Effect of Loss Design. To assess the effect of loss formulation, we compared models trained with only the binary cross-entropy loss $\mathcal{L}_{\text{bce}}$ against those using the combined

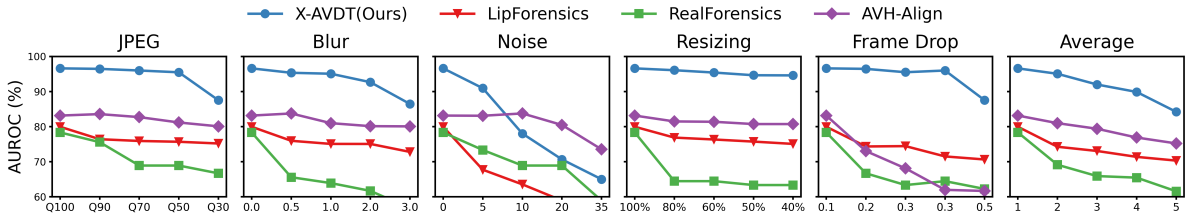


Figure 5. **Robustness against unseen corruptions.** AUROC (%) across five severity levels. Per corruption scales are shown on the x–axes. Average denotes the mean AUROC across all corruptions at each severity level.

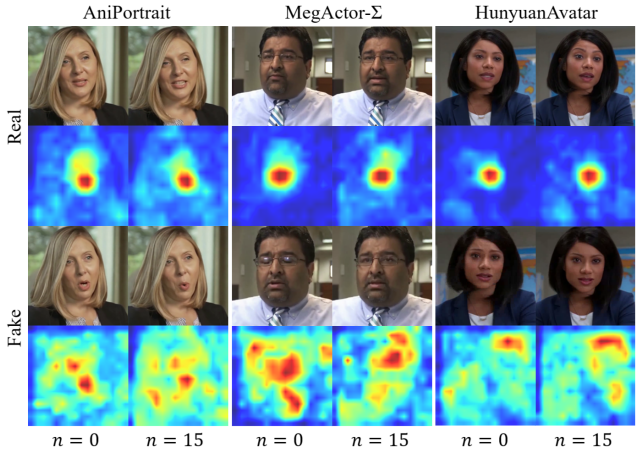


Figure 6. **Grad-CAM visualizations.** Red activation indicates regions where our model focuses most (i.e., pixels that make a strong positive contribution to the predicted class), while cooler colors denote weak or no contribution.

objective $\mathcal{L}_{\text{bce}} + \mathcal{L}_{\text{tri}}$. The triplet loss $\mathcal{L}_{\text{tri}}$ serves as an auxiliary metric-learning loss, providing an inductive bias toward a more discriminative class structure in the embedding space that complements the classification signal. See Table 7b for quantitative results.

Robustness of In-the-Wild Perturbation Attack. We trained X-AVDT on MMDF without augmentation and evaluated robustness on unseen corruptions with five severity levels. For comparison, all baselines used their official pretrained checkpoints. We report AUROC (%) values in Figure 5. X-AVDT outperformed prior detectors across severity levels, with smaller performance decay than comparative methods under high-frequency suppressing distortions (JPEG/Blur), additive noise, and scale changes due to resizing, as well as under temporal disruptions from frame dropping. Detailed information can be found in the supplementary material.

6.5. Discussion

To understand how our detector treats real versus forged videos, we applied Grad-CAM [74] to our detector and visualized activation maps on samples from three representative generators (Figure 6). Grad-CAM highlights where the model grounds its decision in each frame, and these activation maps make the effect of our design explicit. Our method concentrated activation on articulatory regions and maintained this focus across frames in real videos. In contrast, forgeries elicited scattered, multi-focal responses, revealing the absence of consistent audio–visual cues. This behavior was consistent across all generators, indicating that our detector exploits audio–visual consistency as a general cue, which in turn explains its generalization beyond any single model's artifacts. See the supplementary material for details.

7. Conclusion

We introduce X-AVDT, a simple, robust, and generalizable detector that probes internal audio–visual signals in pretrained diffusion models via DDIM inversion. Our approach fuses two complementary representations: a video composite ϕ that surfaces reconstruction-based discrepancies and an AV cross-attention feature ψ that encodes speech–motion synchrony. Empirically, we demonstrate that probing intermediate diffusion features during inversion yields more discriminative and better calibrated signals than relying solely on end point reconstructions. Evaluated on MMDF and external benchmarks, X-AVDT delivers consistent improvements over prior methods and transfers well to unseen generators, achieving superior performance on standard datasets and under perturbed image conditions. We also present MMDF, an audio–visual deepfake benchmark curated for cross-generator generalization and robustness studies. We hope MMDF serves as a strong benchmark suite for advancing detector generalization and real-world robustness, while X-AVDT provides a solid baseline and diagnostic probe for future work on audio–visual, generator-internal cues.

Limitations and Future Work Despite strong accuracy and cross-dataset robustness, our method incurs a high computational cost, reflecting the inherent expense of inversion. For a 16 frame clip, a full DDIM inversion and reconstruction process with a 40 timestep schedule requires approximately one minute end-to-end. Current detectors also remain imperfect in non-speech segments or multi-speaker scenes, because the approach relies on speech-driven features. A few promising directions include supplementing our representation with unimodal back-off strategies for weak or missing speech and with speech-agnostic correspondence cues beyond phonetics, and pursuing lightweight inversion via distillation.

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